

Arihant Jain

Senior Backend Engineer · Distributed Systems & Platform Infrastructure · Python · AWS

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EXPERIENCE

Perfios (Karza Technologies)

Senior Software Engineer

New Delhi

Apr 2024 – Present

- Delivered **\$700K+** in cumulative cost savings by replacing a fixed-order source chain with threshold-based source switching and caching, reducing third-party vendor dependency by 55%, and by shutting down redundant infrastructure.
- Designed and rolled out a shared framework that standardized common service code across **200+** production APIs used by multiple teams, so changes that once meant redeploying every API now ship as one package update.
- Owned multi-region active-passive disaster recovery across Mumbai and Hyderabad for the KYC platform, which serves **5M+** daily requests at 99.9%+ availability.
- Created hourly, daily, and event-triggered monitoring that checks traffic, latency, and errors against seven-day trends and skips planned maintenance, proactively flagging problems and bringing incidents down from **12–13 a week to 2–3**.
- Developed a Redis-backed session library that uses distributed locks so concurrent Lambda runs share one login instead of competing for it, eliminating **35%** of session-related failures and recovering 15K–30K requests a day that previously failed.
- Built proxy-routing infrastructure for 120+ APIs that tracks per-IP health across in-house and third-party pools and pauses failing IPs for a cooldown, raising request success rate by **21%** while proxy incident calls fell from 9–11 a week to 1–2.
- Introduced triage and debugging practices for junior and support engineers, cutting incident triage SLA **from 3 days to 3–4 hours**, and strengthened delivery by implementing multi-region GitLab CI/CD with SAST and parallel runners.

Software Engineer I

Oct 2022 – Apr 2024

- Engineered a queue-based Celery service on AWS Fargate, backed by Redis, to replace a Lambda workflow, **tripling throughput**; session reuse, caching, and rate limits then enabled full retirement of an external vendor.
- Led migration of **250+** KYC and utility APIs from Python 3.9 to 3.12, containerizing over 90% for cloud and on-premises deployment with zero production rollbacks.
- Reduced P99 latency from **8s to 3s** on critical legacy client-facing KYC APIs.
- Developed a vendor-billing and invoicing service used across multiple products, improving billing accuracy by **25%**.
- Shipped **30+ APIs within six months** and maintained 100+ production APIs spanning 300+ verification and data integrations.

InTimeTec

Software Engineer

Jaipur

Mar 2021 – Sep 2022

- Converted a manufacturing client's .NET desktop operations tool into an Angular web platform, growing its user base from a handful of on-site staff to **~70**; went on to build 6 new modules from scratch.
- Owned end-to-end delivery and client communication for a 4-engineer pod, shortening deployment cycles by **44%** and increasing test coverage by 54% through CI/CD and automated testing.

TECHNICAL SKILLS

Languages	Python, JavaScript (ES6+), SQL
Backend & System Design	FastAPI, Node.js, REST APIs, Pydantic, Microservices, Event-Driven Architecture
Data & Messaging	Redis, MongoDB, PostgreSQL, Celery, Kafka, RabbitMQ, Amazon SQS
Cloud & Delivery	AWS Lambda, Fargate, ECS, S3, EC2, CloudFormation, Docker, GitLab CI/CD, CloudWatch Logs Insights
Testing & Security	Pytest, Playwright, Selenium, SAST, VAPT

PROJECTS

Distributed Quota-Metering Engine | FastAPI, Redis Lua, MongoDB, Docker Compose

github.com/Arihant416/quota-meter

- Built a per-organization, per-feature quota service where a single Redis Lua script handles validation, idempotency, deduction, and denial recording atomically, with all-or-nothing batch requests, idempotent refunds, and monthly resets without scheduled jobs.
- Tested with 10,000 concurrent requests and **zero over-serving**; retries with the same idempotency key consumed quota only once.

EDUCATION

Swami Keshvanand Institute of Technology

B.Tech, Information Technology

Jaipur

Aug 2017 – Jun 2021